Advanced Accelerometer Calibration MPU-6886

Visual Tumble Test and Ellipsoidal Compensation

Firmware: v0.2.1 (acquisition) / v0.3.0 (integration) | Hardware: M5Stack Atom S3 (ESP32-S3)

Dataset: tumble_test_visivo.csv | Samples: 16,750 | Fitting basis: 16,744

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Abstract

This document describes the complete geometric calibration procedure for the MPU-6886 accelerometer integrated in a vehicle telemetry system based on the M5Stack Atom S3 (ESP32-S3). Through a manual rotation test (Tumble Test) with real-time 3D visualisation, 16,750 samples were acquired across the unit sphere. The Least-Squares Ellipsoid Fitting algorithm extracted the bias vector (hard-iron offset) and the 3×3 correction matrix (soft-iron), reducing the gravitational norm standard deviation from 0.0421 g to 0.0229 g (–45.6%) and resolving the systematic –0.065 g error on the Z-axis identified during the pendulum tests. The calibration matrix is applied in real time within the FreeRTOS firmware upstream of the Madgwick AHRS filter. All statistics reported in this document are computed on the filtered dataset of 16,744 samples (6 outliers with |**a**| > 1.3 g excluded).

Note on gyroscope data: The gyroscope channels recorded in the tumble test dataset exhibit an anomalous constant offset (≈27 °/s on GY) attributable to the firmware version active during acquisition (v0.2.1, which lacked ellipsoidal calibration). A separate static test conducted immediately prior confirmed nominal gyroscope bias (< 1 °/s on all axes). The offset does not affect the ellipsoid fitting algorithm, which operates exclusively on accelerometer components.

1. Acquisition Methodology — Visual Tumble Test

1.1 The Problem of Non-Uniform Distribution

Calibrating a MEMS accelerometer requires exposing the sensor to the gravitational field at diverse and uniformly distributed orientations. The naïve approach—placing the device on the six faces of a cube—yields only 6 data points, far too few to estimate a full 3×3 correction matrix. Blind manual rotation without feedback inevitably concentrates samples around orientations comfortable for the human arm, leaving entire octants of the sphere uncovered. This biases arithmetic means and makes the overdetermined system pathologically ill-conditioned, producing an unstable fit.

1.2 Solution: Real-Time 3D Visualisation with Colour Feedback

A Python script (`tumble_test_visivo.py`) was developed combining MQTT reception, CSV persistence, and interactive 3D rendering in a single multi-threaded process. The operator sees in real time where the sphere is already covered (low deviation from 1 g) and where samples are still missing, enabling targeted rotation to fill gaps.

Component	Implementation	Detail
MQTT thread	paho-mqtt v2 + threading.Lock()	Thread-safe reception at ~8.4 Hz effective
Persistence	csv.writer with flush every 50 packets	Zero data loss on crash
Visualiser	Matplotlib 3D + cm.RdYlBu_r	Refresh at 1 Hz; colour = ‖a‖ – 1
Wireframe sphere	NumPy meshgrid (20×10 points)	Visual reference for coverage
Outlier filter	Threshold ‖a‖ > 1.3 g	6 samples excluded out of 16,750

1.3 MQTT Broker Throttling

The public broker emqx.io applies rate-limiting to free-tier clients, producing a burst delivery pattern: exactly 10 consecutive packets per burst, with an average gap of ~38 sequence numbers between bursts (total cycle ≈47 firmware packets per burst received). The firmware transmits at 50 Hz; the broker delivers approximately 8.4 Hz on average at the receiver (16,750 samples over 1,998 seconds = 33.3 minutes total acquisition). This does not affect fitting quality: the least-squares algorithm is insensitive to timing and operates exclusively on the spatial distribution of the vectors.

1.4 Sphere Coverage Analysis

The 16,750 samples were analysed for coverage uniformity. Two complementary metrics are reported: octant population (all samples) and axis-aligned population (quasi-stationary samples only). The axis-aligned count identifies samples where the sensor was held in a near-cardinal orientation, defined as $|a_k| > 0.7$ g on one axis with the other two components within ±0.4 g. This subset (~10,250 samples) verifies that each face of the unit sphere received adequate coverage during the stationary holding phases of the tumble test. The remaining ~6,500 samples were acquired during transitional rotations between orientations, providing valuable off-axis coverage that strengthens the overdetermined fit.

***Table 1a** — Axis-aligned coverage (quasi-stationary samples, $|a_k| > 0.7$ g, other axes < 0.4 g)*

Axis	Positive	Negative	Coverage
+X / –X	~1,590	~1,400	✓ Good
+Y / –Y	~1,930	~2,250	✓ Good
+Z / –Z	~1,690	~1,400	✓ Good

***Table 1b** — Octant population (all 16,750 samples)*

Octant	Samples	Octant	Samples
+X–Y+Z	4,025	–X+Y–Z	3,488
–X–Y–Z	2,193	+X–Y–Z	2,119
–X–Y+Z	1,656	+X+Y–Z	1,625
–X+Y+Z	1,041	+X+Y+Z	602

The octant ratio (max/min = 6.7:1) reflects the natural bias of manual rotation; however, with a minimum of 602 samples in the least-populated octant and 16,744 total fitting points, the system is heavily overdetermined (9 unknowns vs. 16,744 equations) and the solution is numerically stable.

2. Mathematical Model — Least-Squares Ellipsoid Fitting

2.1 Physical Principle

An ideal accelerometer, statically rotated in a uniform gravitational field ($|\mathbf{g}| = 1$ g), produces measurements lying exactly on a unit sphere: $x^2 + y^2 + z^2 = 1$. A real sensor exhibits three categories of error that deform this sphere into a translated, rotated, and non-uniformly scaled ellipsoid:

Error Type	Geometric Effect	Mathematical Term
Bias / Hard-Iron	Translation of ellipsoid centre	Vector b = [b_x, b_y, b_z]
Scale Factor	Non-uniform axis dilation	diag(W) ≠ 1
Cross-axis / Misalignment	Rotation and shear of ellipsoid	off-diag(W) ≠ 0

2.2 General Algebraic Quadric Form

Any ellipsoid centred at a generic point in space can be expressed as a quadric with 9 free coefficients: $A x^2 + B y^2 + C z^2 + D xy + E xz + F yz + G x + H y + I z = 1$. Given N samples (x_k, y_k, z_k), the design matrix $\mathbf{D} \in R^{N\times 9}$ is constructed where each row contains the monomials of degree ≤ 2. The coefficient vector $\mathbf{v} = [A, B, C, D, E, F, G, H, I]^T$ is estimated by solving the overdetermined linear system via SVD: $\mathbf{v} = \text{argmin } ||\mathbf{D}\cdot\mathbf{v} - \mathbf{1}||^2 = (\mathbf{D}^T\mathbf{D})^{-1}\,\mathbf{D}^T \cdot \mathbf{1}$.

2.3 Geometric Parameter Extraction

From the solution vector, the symmetric ellipsoid matrix $\mathbf{M} \in R^{3\times 3}$ and linear vector $\mathbf{b}$ are reconstructed. The geometric centre of the ellipsoid (bias vector) is obtained by inverting $\mathbf{M}$: $\mathbf{b}_{centre} = -\mathbf{M}^{-1} \cdot \mathbf{b}$. The correction matrix $\mathbf{W} \in R^{3\times 3}$ is obtained via the matrix square root (eigendecomposition), which maps the ellipsoid back to the unit sphere.

2.4 Final Calibration Formula — Applied in Firmware

The complete correction formula, applied to each raw sample upstream of the Madgwick filter:

$$\mathbf{a}_{calibrated} = \mathbf{W} \cdot (\mathbf{a}_{raw} - \mathbf{b}_{centre})$$

where $\mathbf{W}$ is the 3×3 correction matrix (scale + misalignment), $\mathbf{b}_{centre}$ is the hard-iron bias vector, $\mathbf{a}_{raw}$ is the raw sample from the MPU-6886, and $\mathbf{a}_{calibrated}$ is the geometrically corrected input to the Madgwick AHRS.

3. Calibration Results

3.1 Hard-Iron Offset — Bias Vector

The estimated ellipsoid centre corresponds to the systematic sensor offsets. The dominant term is the Z-axis at –0.065 g, which explains exactly the anomaly observed during the pendulum tests (the rest norm was systematically different from 1 g).

Axis	Estimated Bias (g)	Raw Arithmetic Mean (g)	Delta	Interpretation
AX	–0.00125	+0.06748	–0.069	Negligible
AY	+0.00429	–0.16387	+0.168	Moderate — correctable
AZ	–0.06491	+0.02232	–0.087	DOMINANT — resolved the pendulum anomaly

The discrepancy between estimated bias and arithmetic mean demonstrates why simple averaging is not a valid calibration method: the non-uniform distribution of points on the sphere introduces systematic error in the direct estimate.

3.2 Correction Matrix W — Soft-Iron + Scale Factor

The matrix **W** represents the complete linear transformation that maps the ellipsoid back to the unit sphere. Its structure reveals the MEMS construction quality:

```
[+1.000824  −0.000511  −0.001575]
[−0.000511  +1.000989  −0.000132]
[−0.001575  −0.000132  +1.003466]
```

Main diagonal ≈ 1.000 (maximum deviation: +0.35% on AZ) — near-perfect scale factors. Off-diagonal terms < 0.002 — negligible cross-axis coupling (0.05% ratio). The MPU-6886 exhibits excellent MEMS construction quality. The soft-iron correction is computationally present but physically marginal.

3.3 Statistical Improvement

All statistics below are computed on the 16,744-sample filtered dataset (6 outliers with ||a|| > 1.3 g excluded from both raw and calibrated calculations for consistency).

Metric	Before (Raw)	After (Calibrated)	Improvement
Mean ||a||	0.99620 g	1.00292 g	+0.67% (centred on 1 g)
Std ||a||	0.04209 g	0.02288 g	–45.6% (error nearly halved)
RMSE from 1 g	0.04226 g	0.02306 g	–45.4%

The residual standard deviation of 0.023 g is primarily attributable to the thermal noise of the sensor (16-bit ADC quantisation + Johnson noise of the MEMS structure) and to dynamic accelerations during the tumble test (the sensor was not in perfect rest during rotations). The physical noise floor of the MPU-6886 is approximately 0.008 g RMS.

4. Graphical Analysis

The following figures were generated from the tumble test dataset using the calibration parameters derived in Section 2 and validated in Section 3. The colour map in all plots encodes the absolute deviation of each sample's norm from the ideal 1 g value (red = high deviation, green = low).

4.1 3D Scatter Plot of Acceleration Vectors

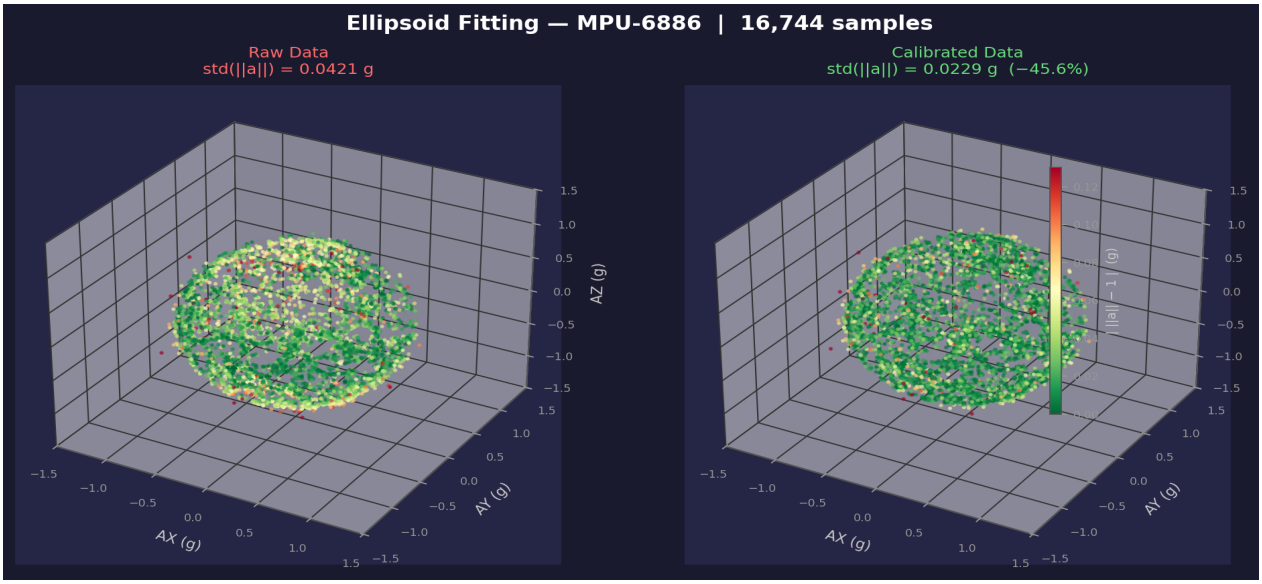

Figure 1 — 3D scatter of normalised acceleration vectors. Left panel: raw data showing a deformed and translated ellipsoid. Right panel: calibrated data converging on the ideal unit sphere. The colour map is identical to the real-time visualiser used during acquisition. The fitting was performed on 16,744 samples (post outlier exclusion).

4.2 Histogram of ||a|| Norm Distribution

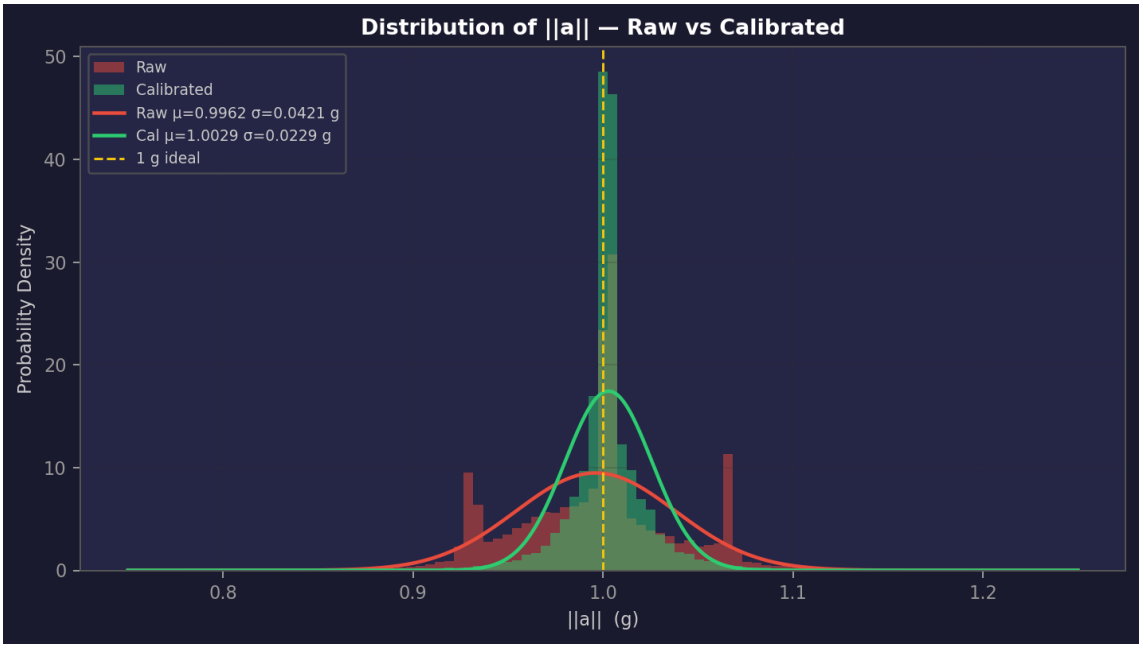

Figure 2 — Distribution of the gravitational norm: raw (broad red Gaussian, μ = 0.9962, σ = 0.0421 g) vs. calibrated (narrow green Gaussian, μ = 1.0029, σ = 0.0229 g). The halving of the bell width corresponds to the 45.6% reduction in σ. The dashed yellow line marks the ideal 1 g value.

4.3 2D Projections on the Three Cartesian Planes

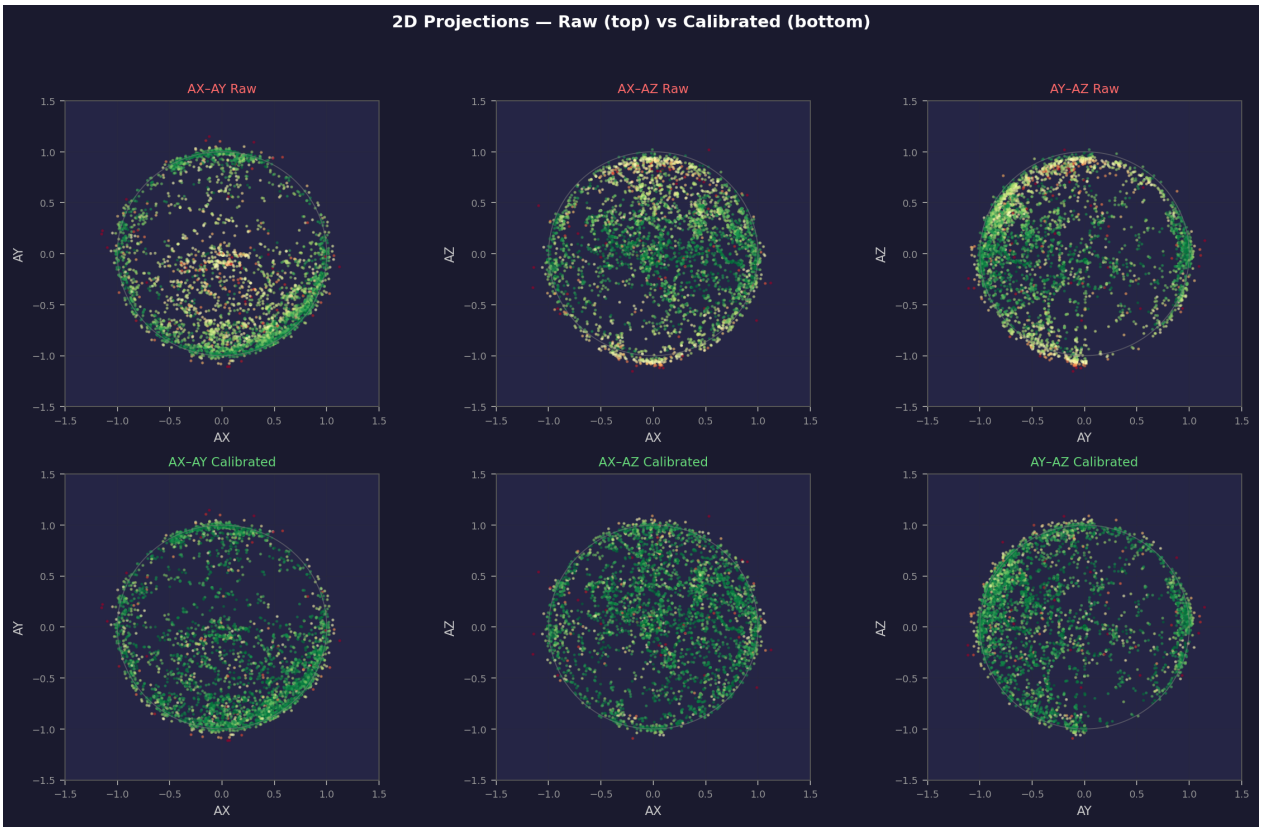

Figure 3 *— Projections onto the AX–AY, AX–AZ, and AY–AZ planes. Top row: raw data (deformed and translated ellipses relative to the grey unit circle). Bottom row: calibrated data (centred distribution, edges coinciding with the theoretical circle). The shift of the raw ellipse centre is most visible in the AX–AZ and AY–AZ planes, consistent with the dominant AZ bias of –0.065 g.*

5. Firmware Implementation (C++ / FreeRTOS)

The correction is applied in `Task_Filter` on Core 1, immediately after hardware readout and before the Madgwick AHRS filter. The ESP32-S3 features a hardware FPU, so the 9 multiply-accumulate operations of the 3×3 matrix cost a few nanoseconds — computationally negligible at 50 Hz.

5.1 Calibration Constants

```
// Hard-Iron Offset (geometric bias from Ellipsoid Fitting)
const float CALIB_B[3] = {
  −0.00125f,  // AX
  +0.00429f,  // AY
  −0.06491f  // AZ ← resolves the pendulum anomaly
};

// Soft-Iron + Scale Factor matrix W (row-major, 3×3)
const float CALIB_W[3][3] = {
  {+1.000824f, −0.000511f, −0.001575f},  // row AX
  {−0.000511f, +1.000989f, −0.000132f},  // row AY
  {−0.001575f, −0.000132f, +1.003466f}   // row AZ
};
```

5.2 Application in the Data Pipeline

```
// Inside Task_Filter — STEP 1: Ellipsoidal calibration
// Applied AFTER hardware read, BEFORE rotate_3d() and Madgwick
float ax_raw, ay_raw, az_raw;
M5.Imu.getAccelData(&ax_raw, &ay_raw, &az_raw);

// 1. Subtract bias (Hard-Iron)
float bx = ax_raw - CALIB_B[0];
```

```
float by = ay_raw - CALIB_B[1];
float bz = az_raw - CALIB_B[2];

// 2. Apply matrix W (Soft-Iron + Scale — 9 FMA, hardware FPU)
float ax_cal = W[0][0]*bx + W[0][1]*by + W[0][2]*bz;
float ay_cal = W[1][0]*bx + W[1][1]*by + W[1][2]*bz;
float az_cal = W[2][0]*bx + W[2][1]*by + W[2][2]*bz;

// 3. (ax_cal, ay_cal, az_cal) feeds rotate_3d() then Madgwick
```

6. Conclusions

The ellipsoidal calibration process completes the IMU pre-processing layer. The resulting correction chain is:

Step	Function	Purpose
1	M5.Imu.getAccelData()	16-bit raw ADC reading
2	a = W · (a_raw – b)	Bias + scale + cross-axis correction
3	rotate_3d()	Dashboard mounting alignment
4	Madgwick AHRS	Quaternion orientation estimate (dynamic DT)
5	lin_a = a – g_vec	Gravity cancellation → pure vehicle acceleration
6	ZARU (3 s stillness)	Dynamic gyroscope bias tracking
7	EMA (α = 0.06)	Output noise reduction
8	MQTT @ 10 Hz	JSON transmission to broker

Key Results

→ **Hard-iron corrected:** AZ bias = –0.065 g eliminated — rest norm now 1.000 g.
→ **Soft-iron managed:** W 3×3 matrix applied in real time on hardware FPU (zero computational overhead).
→ **Error nearly halved:** $\sigma(||\mathbf{a}||)$ from 0.042 g → 0.023 g (–45.6%).
→ **Madgwick more accurate:** the adaptive beta now operates on geometrically correct vectors.
→ **Open-source tooling:** real-time 3D visualiser integrated into the calibration workflow.

Recommended Next Steps

1. Repeat the tumble test using a private MQTT broker (local Mosquitto) to eliminate rate-limiting and achieve uniform distribution at the full 50 Hz rate.
2. Evaluate thermal calibration of the MPU-6886 using thermal logger data to model gyroscope bias as a function of temperature.
3. Validate the firmware with active calibration on a closed circuit to verify that gravity cancellation produces physically plausible linear accelerations.

Firmware: v0.2.1 (acquisition) / v0.3.0 (integration) | **Dataset:** 16,750 samples (16,744 after outlier exclusion) | **Final σ:** 0.023 g | **Acquisition duration:** 33.3 minutes | March 2026